

CS101

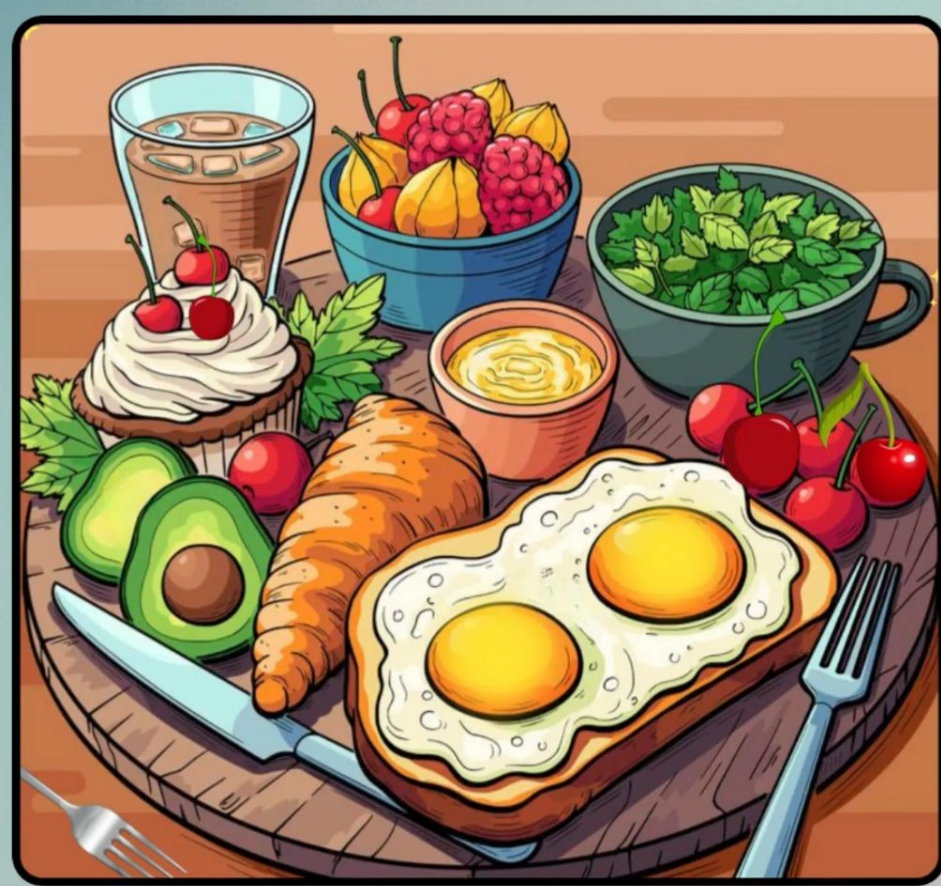
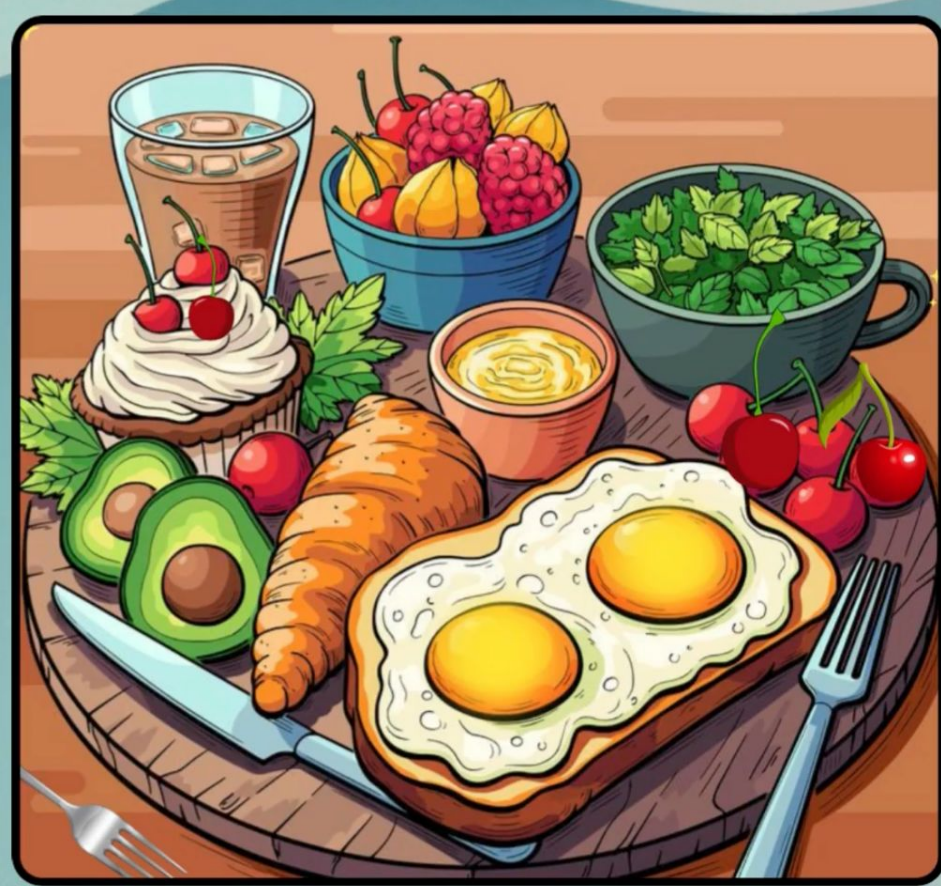
Variables & Arithmetic

1/29

Agenda

1. Quiz2
2. Spot the Differences
3. Demo Some CS Vocab
4. Lab3 + YouTube Tutorials
5. Windows OS Tips + Touch Typing

Spot the Differences



<https://www.youtube.com/watch?v=vk6jwZ--caE>

```
public class practice {  
    public static void main(String[] args) {  
        System.out.println("Example Print Statements:\n");  
    }  
}
```

```
public class practice {  
    public static void main(String[] args) {  
        System.out.print("Example Print statements:");  
    }  
}
```

```
public class practice {  
    public static void main(String[] args) {  
        int age = 80;  
        double height = 170.5;  
        String address = "1615 Ripley Ln";  
  
        System.out.println("Age: " + age);  
        System.out.println("Height: " + height);  
        System.out.println("Address: " + address);  
    }  
}
```

```
public class Practice {  
    public static void main(String[] args) {  
        int age = 80;  
        double height = 170.5  
        String address = "1615 Ripley Ln";  
  
        System.out.println("Age: " + age);  
        System.out.println("Height: " + height);  
        System.out.println("Address: " + address);  
    }  
}
```

CS Vocab (demo)

Declaration: define variable name and type

Assignment: assign a value to a variable

Initialization: set variable's initial value

Increment: increase variable (usually by 1)

Decrement: decrease variable (usually by 1)

cs50 Duck AI

File Edit Selection View Go ... practice.java — 67756774 [Codespaces: bug-free space lamp] — Visual Studio Code



CS50 DUCK DEBUGGER

ddb
Quack. I am CS50's duck debugger (ddb), an experimental AI for [rubberducking](#). Quack quack. My replies might not always be accurate, so always think critically and let me know if you think that I've erred. Conversations are logged. Quack quack quack.

Ask a question



practice.java ×

```
1 public class practice {  
2     public static void main(String[] args) {  
3     }  
4 }  
5  
6
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

\$ □

Lab3 & YouTube Tutorials

Java Primitive Types

Type	Size	Range	Example	Description
byte	8-bit	-128 to 127	byte b = 100;	Integer
short	16-bit	~±32 thousand	short s = 1500;	Integer
int	32-bit	~±2.1 billion	int i = 42000;	Integer
long	64-bit	~±9.2 quintillion	long l = 99999L;	Integer
float	32-bit	~±3.4E+38 (6–7 digits)	float f = 3.14f;	Decimal
double	64-bit	~±1.8E+308 (15–16 digits)	double d = 3.14159;	Decimal
char	16-bit	Unicode characters	char c = 'A';	Character
boolean	1-bit	true or false	boolean b = true;	Boolean

We (mostly) use

- int
- double

Lab3 Questions

HW1 Flipbook

Due Fri Feb 6 at 11:59pm

Next Class 2/3

Before Next Class:

- First submissions of Lab1/2/3
- Start HW1

Next Class

1. Folder Structures?
2. Input from user
3. Start Lab4

- **Bring Headphones every class!**
- I'll bring food if everyone comes to an office hours or schedule a 1on1 **before Tuesday 2/3**